Custom Rescue Node Pinout Guide (V2)

> **Optimal Layout:** This layout fully integrates your Tactile Button and Piezo/LED using clever hardware routing, allowing exactly 11 components to perfectly match the Seeed Studio XIAO ESP32C3's 11 GPIO pins.

Overall Pin Assignment Summary

XIAO Pin	ESP32-C3 GPIO	Component	Function
D9	`GPIO 9`	**Tactile Button**	Connect external button to D9 & GND
D8	`GPIO 8`	**SX1262 LoRa**	SPI SCK (Clock)
D10	`GPIO 10`	**SX1262 LoRa**	SPI MISO
D7	`GPIO 20`	**SX1262 LoRa**	SPI MOSI
D6	`GPIO 21`	**SX1262 LoRa**	SPI NSS (Chip Select)
D5	`GPIO 7`	**SX1262 LoRa**	DIO1 (Interrupt)
D4	`GPIO 6`	**SX1262 LoRa**	BUSY
D3	`GPIO 5`	**MPU6050**	I2C SDA
D2	`GPIO 4`	**MPU6050**	I2C SCL
D1	`GPIO 3`	**NEO-6M GPS**	RX (Connects to GPS TX only)
D0	`GPIO 2`	**Buzzer/LED**	Piezo Buzzer & Status LED (Parallel)

Special Hardware Routing Instructions

Component	Special Routing
SX1262 RESET	Do not connect to a GPIO pin. Connect the LoRa RESET pin directly to the XIAO's hardware `EN` or `RST` pin so they reboot together.
NEO-6M GPS TX	Do not connect the ESP32's TX back to the GPS RX. Meshtastic will listen passively on its RX pin.

Power Connections

XIAO Pin	Connects To
3V3	SX1262 VCC, GPS VCC, MPU6050 VCC
GND	SX1262 GND, GPS GND, MPU6050 GND, External Button, Buzzer/LED Ground

Generated for Meshtastic Custom Firmware ESP32-C3 (Seeed XIAO)